# Project – High Level Design

# On

# Agriculture Quiz/ Tutor Bot

Course name: Generative ai

**Institution name:** Medicaps University – Datagami skill based course

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# 1. Introduction

The Agriculture Quiz/Tutor Bot is a Generative AI-based educational system designed to help users learn agriculture concepts through interactive quizzes and intelligent feedback. The system uses Large Language Models (LLMs) to understand the meaning of user answers and evaluate them accurately instead of checking only fixed keywords. It provides explanations for wrong answers and helps users improve their knowledge in topics like farming methods, soil care, crop production, and irrigation techniques. The main aim of this project is to make learning more engaging, personalized, and effective.

## 1.1. Scope of the document.

This document explains the **high level design** of the Agriculture Quiz/Tutor Bot, including how the system is built and how different components work together. It covers quiz generation, AI-based answer evaluation, system architecture, user interaction, APIs, and database storage. It also describes how the frontend, backend, AI module, and database communicate to provide accurate results and feedback.

## 1.2. Intended Audience

This document is useful for:

- **Developers** – to understand system structure and implementation
- **Designers** – to understand interface and user experience flow
- **AI/ML Engineers** – to understand how Generative AI and LLM are integrated
- **Students** – to learn how an AI-based project is designed
- **Project Evaluators** – to review the technical working of the system
- **Technical Users** – to understand architecture, APIs, and workflow of the project

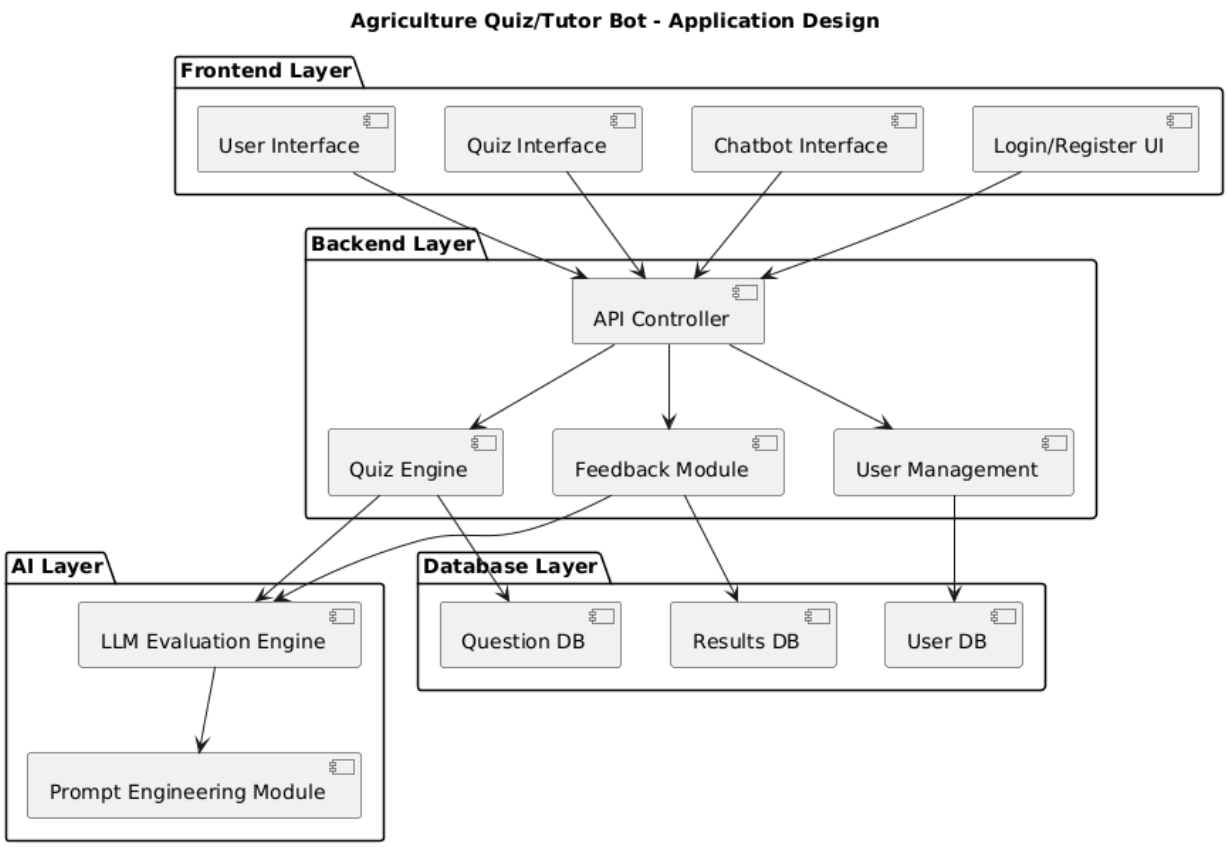
## 1.3. System overview

- The **Agriculture Quiz/Tutor Bot** is an AI-powered learning application for agriculture education.
- It conducts quizzes on topics such as **Farming Tips, Soil Management, Crop Production, and Irrigation Techniques**.
- Users can select a topic and answer questions through a simple and user-friendly interface.
- The system uses **Generative AI (LLM)** to evaluate answers based on meaning, not just keywords.
- It provides **scores along with detailed explanations** for incorrect answers.
- The system identifies **weak areas** of the user and suggests improvements.
- Chatbot interaction allows users to learn in a conversational and interactive way.
- The platform combines **quiz system + AI evaluation + personalized feedback** to improve learning experience.
- It helps users understand concepts clearly and improve knowledge step-by-step.

# 2. System Design

## 2.1. Application Design

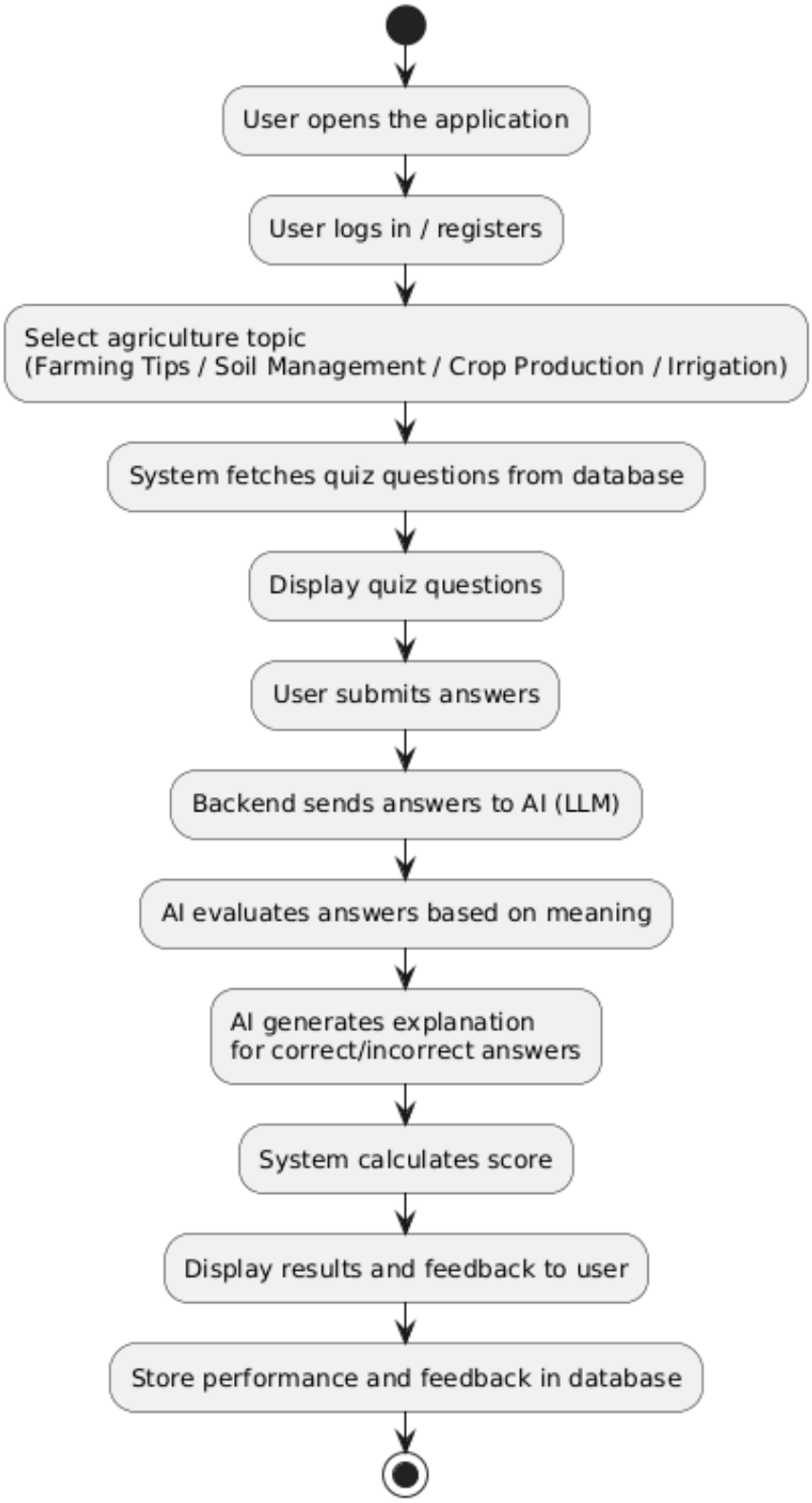
- The system follows a **Client-Server architecture integrated with Generative AI** to provide intelligent quiz evaluation.
- It is divided into **four main layers**, each performing a specific function:
  - **Frontend Layer** – Provides the user interface where users can register, login, select topics, attempt quizzes, and interact with the chatbot.
  - **Backend Layer** – Handles the main logic of the application such as managing quizzes, processing answers, generating feedback, and handling user data.
  - **AI Layer** – Uses **Large Language Model (LLM)** to evaluate answers based on meaning and generate detailed explanations.
  - **Database Layer** – Stores important information like user profiles, quiz questions, answers, scores, and feedback.
- This layered design makes the system **organized, scalable, and easy to maintain**.




## 2.2 Process Flow.

- First, the **user logs into the system** using login credentials.
- The user **selects a topic** related to agriculture such as farming tips or soil management.
- The system **retrieves quiz questions** from the database according to the selected topic.
- The user **submits answers** through the quiz interface.
- The **backend sends the answers to the AI model** for evaluation.
- The **LLM checks the correctness** of answers using semantic understanding.
- The AI **generates explanations** for correct and incorrect answers.
- The system **displays score, explanations, and feedback** to the user.
- The performance data is **stored in the database** for future learning improvement.

**Agriculture Quiz/Tutor Bot - Process Flow**



## 2.3. Information Flow

**Input:**

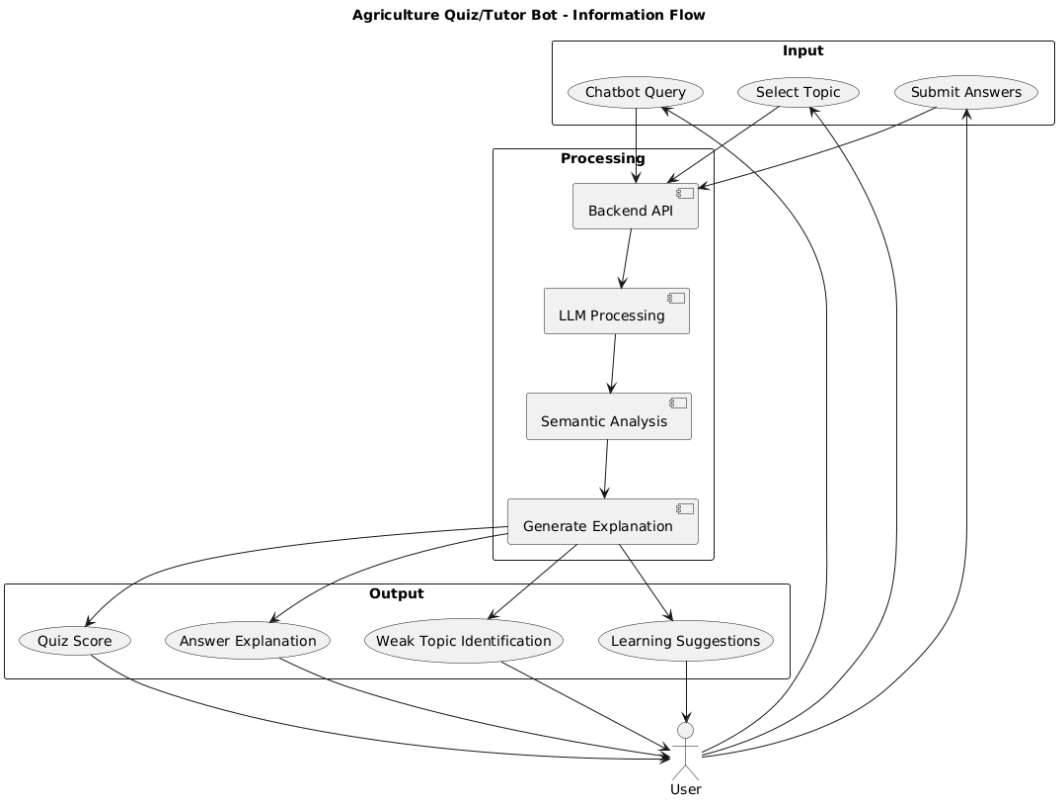
- User enters answers to quiz questions.
- User selects topic for the quiz.
- User interacts with chatbot for help or explanation.

**Processing:**

- Backend APIs receive user input and process it.
- AI model performs semantic analysis to understand the meaning of answers.
- AI generates context-based explanations and feedback.

**Output:**

- Quiz result (correct or incorrect answers).
- Explanation of answers.
- Identification of weak topics.
- Suggestions to improve knowledge.
- Personalized learning recommendations.



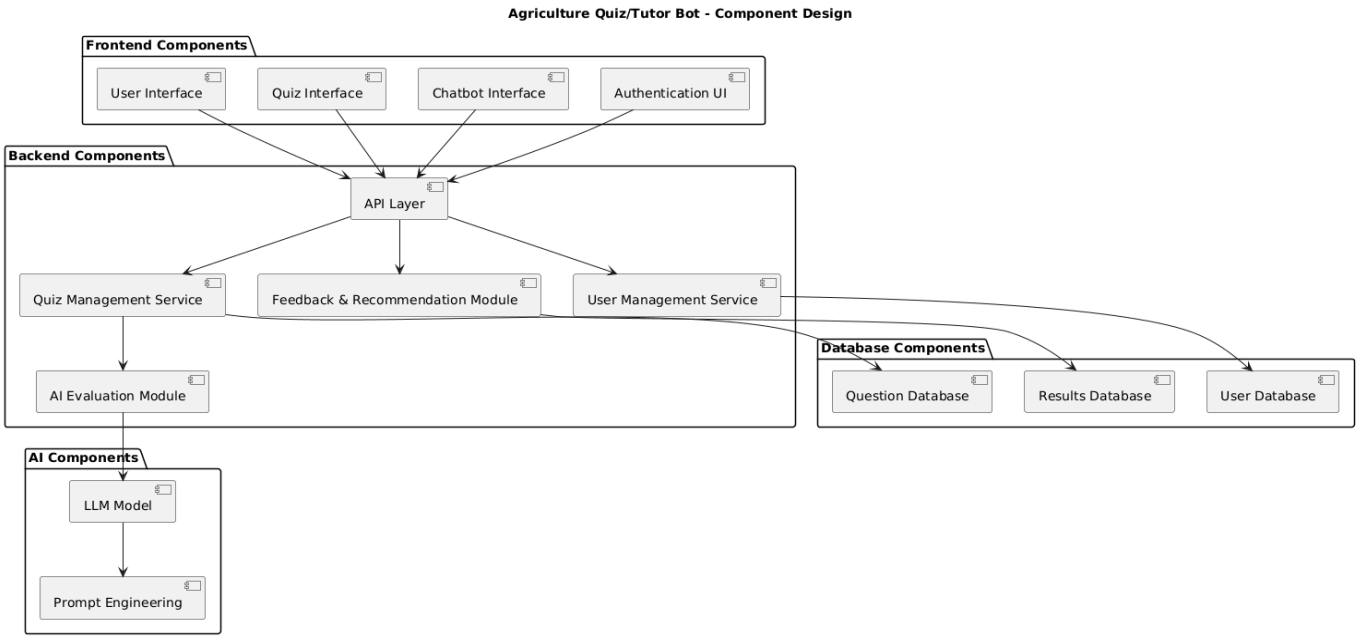
## 2.4.Components Design

**Frontend Components:**

- User Interface (UI)
  Displays quiz questions, options, explanations, and results in a simple format.
- Quiz Interface
  Allows users to select agriculture topics and submit answers easily.
- Chatbot Interface
  Enables users to communicate with AI and receive explanations or suggestions.
- User Authentication UI
  Provides login and registration functionality for secure access.

**Backend Components:**

- API Layer
  Handles communication between frontend and backend services.
- Quiz Management Service
  Controls question selection, quiz sequence, and validation of answers.
- AI Evaluation Module
  Sends user answers to LLM and receives evaluation results.
- Feedback & Recommendation Module
  Analyzes user performance and identifies weak areas.
- User Management Service
  Stores and manages user details, login credentials, and profiles.
- Database Layer
  Stores questions, answers, scores, feedback, and user information securely.




## 2.5. Key Design Considerations

- Accuracy
  Uses AI-based semantic understanding to evaluate answers correctly.
- Scalability
  Modular architecture allows the system to support more users and topics in future.
- Performance
  Optimized APIs ensure fast response time and smooth quiz experience.
- Security
  Secure login system protects user data and privacy.
- User Experience
  Simple interface, easy navigation, and real-time feedback improve learning.
- Flexibility
  System can be expanded to include more agriculture topics or AI features.

## 2.6. API Catalogue.

- **Get Questions API**
  - Endpoint: /getQuestions
  - Method: GET
  - Function: Retrieves quiz questions based on selected agriculture topic.
- **Submit Answer API**
  - Endpoint: /submitAnswer
  - Method: POST

  - Function: Sends user answer to system and returns score and explanation.
- **AI Evaluation API**
  - Endpoint: /evaluate
  - Method: POST
  - Function: Uses LLM to check correctness and generate explanation.
- **User APIs**
  - /login – Allows existing users to access the system.
  - /register – Allows new users to create account.
- APIs enable smooth communication between frontend, backend, and AI module.

# 3.Data Design.

## 3.1. Data Model

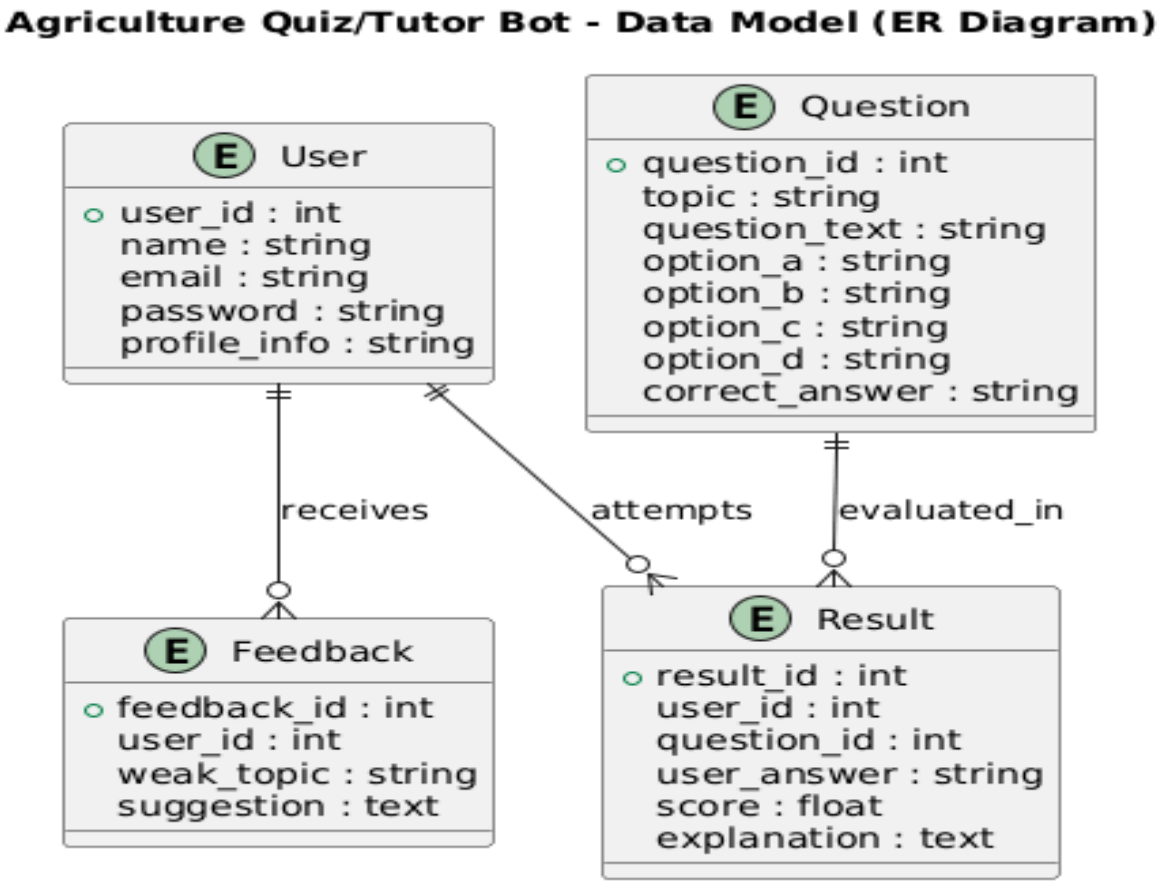
The data model defines how data is structured and stored in the system. The Agriculture Quiz/Tutor Bot mainly uses the following data entities:

- **User Data**
  - User ID
  - Name
  - Email
  - Password
  - Profile details
- **Question Data**
  - Question ID
  - Topic (Farming Tips, Soil, Crop, Irrigation)
  - Question text
  - Options (for MCQ)
  - Correct answer
- **Quiz Result Data**
  - Result ID
  - User ID
  - Question ID
  - User Answer

- Score
- Explanation generated by AI

- **Feedback Data**
  - Weak topics
  - Suggestions for improvement

The data model helps in organizing and managing quiz information efficiently.




## 3.2. Data Access Mechanism

Data access mechanism defines how the system reads and writes data in the database.

- Backend communicates with database using APIs.
- CRUD operations are used:
  - Create – store new users, questions, results
  - Read – fetch quiz questions and user data
  - Update – update scores and feedback

  - Delete – remove unnecessary or old data if required
- Secure authentication ensures only authorized users can access data.
- AI module retrieves question and answer data for evaluation.

### 3.3. Data Retention Policies

Data retention policy defines how long the system stores data.

- User account information is stored securely for long-term use.
- Quiz results are stored to track user performance and progress.
- Old or unused data can be removed after a certain time period.
- Data privacy and security are maintained.
- Backup may be created to prevent data loss.

### 3.4. Data Migration

Data migration refers to transferring data when system changes occur.

- Data can be moved from one database to another if system is upgraded.
- Question bank can be updated or expanded with new agriculture topics.
- Existing user data is preserved during updates.
- Migration ensures system continuity without losing important information.
- Used when shifting from local database to cloud database.

## 4. Interfaces

### 4.1 User Interface (UI)

- The User Interface allows users to interact with the system easily.
- Users can **register, login, select quiz topics, answer questions, and view results**.
- Displays quiz questions, options, scores, and AI-generated explanations.
- Designed to be **simple, clear, and user-friendly** for better learning experience.

### 4.2 Quiz Interface

- Allows users to **select agriculture topics** such as Farming Tips, Soil Management, Crop Production, and Irrigation Techniques.
- Displays **MCQ or text-based questions**.
- Allows users to **submit answers**.
- Shows **next question** and quiz progress.

### 4.3 Chatbot Interface

- Provides **AI-based conversation support**.
- Users can ask doubts related to agriculture topics.
- AI provides **explanations, suggestions, and learning guidance**.
- Makes learning more **interactive and engaging**.

### 4.4 API Interface

- APIs connect **frontend, backend, and AI modules**.
- Handles request and response between system components.
- Important APIs:
  - **/getQuestions** – fetch quiz questions.
  - **/submitAnswer** – submit user answers.
  - **/evaluate** – AI evaluates answers.
  - **/login, /register** – manage users.
- Ensures smooth communication between components.

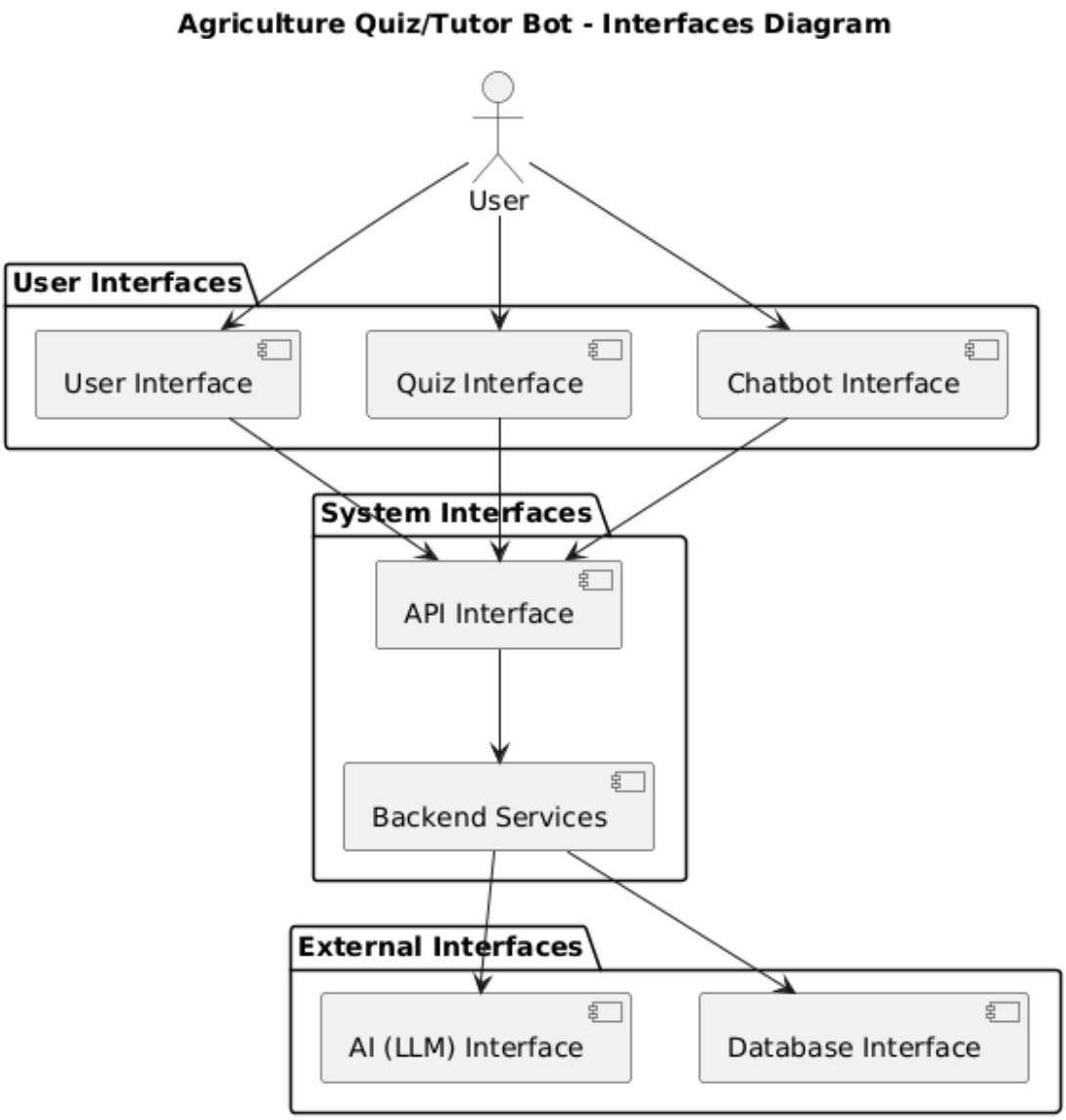
### 4.5 Database Interface

- Manages storage and retrieval of data.
- Stores:
  - User details
  - Quiz questions
  - User answers
  - Scores and feedback
- Backend accesses database using secure queries.
- Ensures **data consistency and security**.

### 4.6 AI (LLM) Interface

- Connects system with **Generative AI model**.
- Sends user answers to AI for evaluation.
- AI returns:
  - correctness result
  - explanation
  - suggestions

- Uses prompt engineering to generate accurate responses.




## 5. State and Session Management

- **Session Management** tracks user activity after login using a unique session ID.
- The system maintains **login state** so only authenticated users can access quizzes and results.
- **Quiz State** stores quiz progress such as selected topic, current question, submitted answers, and score.
- **AI Interaction State** keeps context of user responses to provide accurate explanations and feedback.
- Session data may be stored in **server memory, database, or tokens (JWT)**.

- Security features like **session timeout and authentication** protect user data and prevent unauthorized access.

## 6. Caching

Caching is used to **store frequently used data temporarily** so the system can respond faster.

- Frequently accessed quiz questions can be stored in cache.
- Reduces repeated database calls and improves system speed.
- Improves response time when users request the same topic again.
- Helps reduce load on backend and database.
- Cache can store:
  - Popular quiz questions
  - User session data
  - AI response results (for similar queries)
- Technologies like **Redis or in-memory caching** can be used.
- Cached data is updated periodically to keep information accurate.

## 7. Non-Functional Requirements

Non-functional requirements define how well the system performs rather than what the system does.

- Ensures the system is reliable, secure, fast, and easy to use.
- Helps improve user experience and system quality.
- Important for system performance and scalability.

### 7.1 Security Aspects

Security ensures user data and system resources are protected.

- Secure login and registration using authentication.
- Passwords are stored in encrypted form.
- Only authorized users can access quizzes and results.
- Protects user personal information.
- API access is secured to prevent unauthorized use.
- Prevents data misuse and unauthorized modification.
- Session timeout reduces risk of unauthorized access.

- Data validation prevents incorrect or harmful inputs.

### 7.2 Performance Aspects

Performance ensures the system works efficiently and quickly.

- Fast loading of quiz questions.
- Quick response from AI evaluation.
- Efficient backend processing.
- Ability to handle multiple users at the same time.
- Optimized database queries improve speed.
- Caching helps reduce response time.
- System designed to scale for more users and topics in future.
- Smooth user experience without delays.

## 8. References

References include the sources and materials used to design and develop the Agriculture Quiz/Tutor Bot system.

- OpenAI / Large Language Model (LLM) Documentation – Used for understanding AI-based answer evaluation and explanation generation.
- Generative AI Course Materials – Provided concepts related to AI integration and system design.
- Agriculture Knowledge Sources – Used for creating quiz questions on farming topics such as soil management, crop production, and irrigation.
- Research papers / online tutorials – Helped in understanding chatbot design and AI-based learning systems.
- Technical documentation – Used for API design, database structure, and architecture planning.

These references helped in building an accurate, reliable, and effective AI-based quiz and tutoring system.